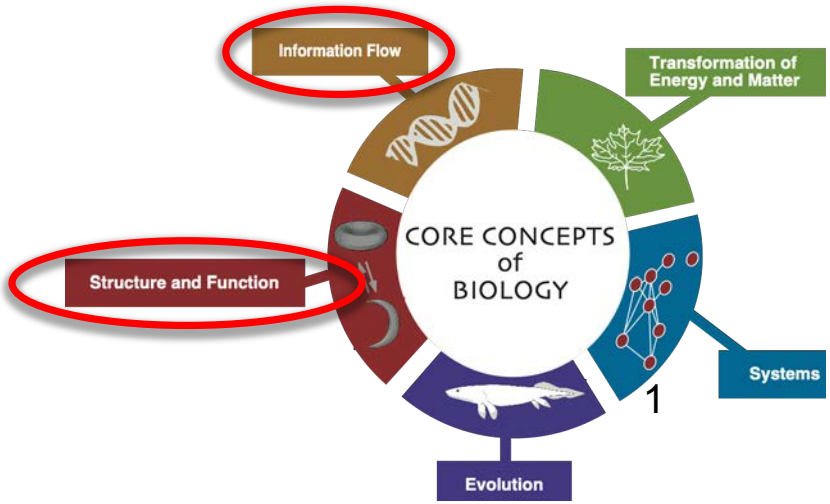
Biol 100: Class #08

Agenda:

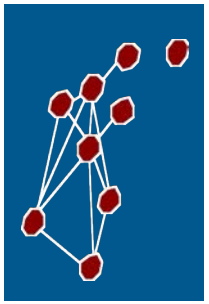
How do cells acquire mutations? What causes mutations? When during cell division do mutations form? What happens to DNA during mitosis? Can peanut butter cause cancer? How do mutagens modify DNA and lead to mutations? How likely is it to pass a mutation to your offspring?

- **Welcome, Agenda, & Announcements**
- **Make NEW Name Tent Card**
- **Pair Discussion: Reflecting on Exam #1**
- **Big Idea: Information Flow in Biology**
- **DNA Replication and mitosis**
- **Discussion: Clicker Question and Unpacking the Peanut Files**
- **Name Tent Card and Homework assignments**



Sally Pasion, Professor
Kimberly Tanner, Professor




Your Class Name Tent

On your name tent…

– explore important dates for upcoming class sessions, quizzes, and exams

– find our emails, student hours, and locations

– at the end of each class session, ***respond to our end of class prompt*** (or record what you most want to remember about that class)

Your Preferred First Name (really big!)

Please leave all name tents in the bin at your table after EVERY class session!



Table Discussion

- **Check-in with your Team Bio 100 table colleagues!**
- **Share one wonderful thing you did since our last class meeting (Exam 1)!**



Pair Discussion

with your shoulder buddy or your team of three:

Reflecting on how you Prepared for Exam 1…

- **Which of your study strategies were most helpful?**
- **Which of your study strategies were the least helpful?**
- **Did you have a study group? Did the person you are talking with right now have a study group? Get each other's information!**
- **How will you change your study habits in preparation for the next exam?**

Person with the shortest hair starts!!





Sally Pasion, PhD

pasion@sfsu.edu

Professor

SFSU Biology Department

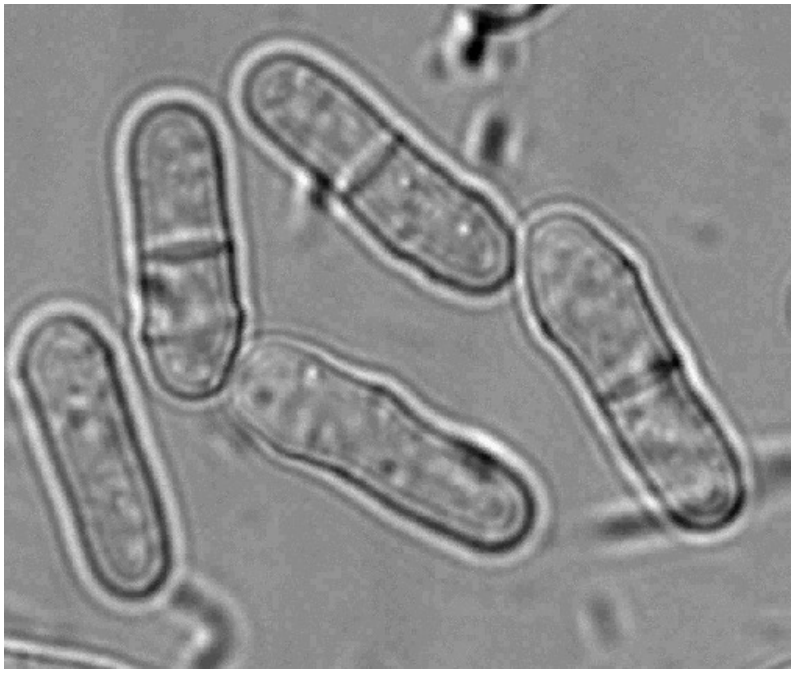
Office Hours on Zoom

Tuesdays, 11:30am – 12:30pm,

Wednesdays, 8:30am-9:30am

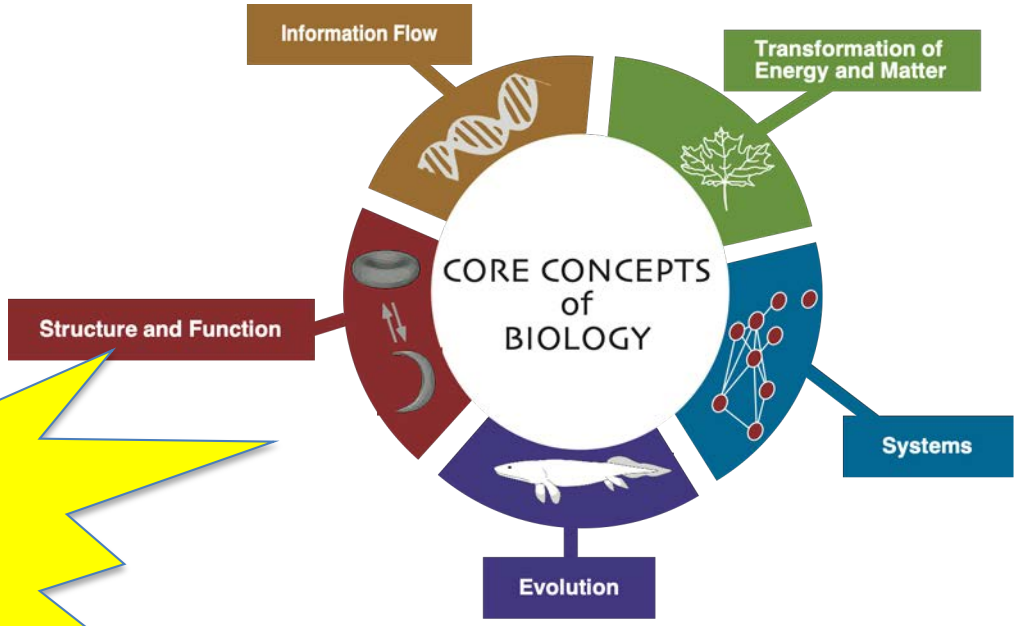
Please make appointment on Navigator:

https://sfsu.campus.eab.com/pal/5tVebeuVMs



Cortes et al., JBC 2012

Big Idea: Five Core Concepts of Biology!



S.I.T.E.S.

Structure and Function

Information Flow

Transformations of Energy & Matter

Evolution

Systems Biology

Big Idea: Five Core Concepts of Biology

Information Flow

Structure and

TS

S.I.T.E.S

Structure and Functi

Information Flow

mations of Energy

Evolution

Systems Biology



PEANUT BUTTER CAN CAUSE CANCER?

Your best friend just came back from traveling abroad. He says that he met people who claimed they got cancer from eating peanuts. He asks you for help to figure out whether this could be true.

Can you explain to your friend whether there could be a link between peanut butter and cancer?

R



INHERITING CANCER

Your cousin develops breast cancer that does not spread to any other tissues. She is thinking about getting pregnant and having children, and she is worried that she might pass on her breast cancer to them. Should she worry?

How, if at all, could it be passed along? How could her breast cancer NOT have any chance of being passed along to her biological children?

N



TWINS + TWINS = TWINS?

A couple astonished diners at their restaurant by appearing to work 21-hour days: opening the restaurant at 6am and still being there to close it at 3am. It turns out that the restaurant in the city of Yiwu in eastern China is actually run by TWO couples – and both the men and the women are identical twins. The twin brothers married the twin sisters three years ago, and they decided to go into business together. Both couples each have a young son, and they look very similar...

What is the chance is that the child on the right in the picture is identical to the child on the left?

F



BABIES FROM A DISH

Your aunt and her husband always wanted a big family, but they just cannot seem to get pregnant.

They've just returned from a reproductive specialist and have lots of questions about IVF–In Vitro Fertilization. They're amazed that it's been around so long, since Louise Brown was the first IVF baby in 1978. They have lots of questions of you, since you have taken college biology.

How would you explain IVF to your auntie?

L



SAVIOR SIBLINGS

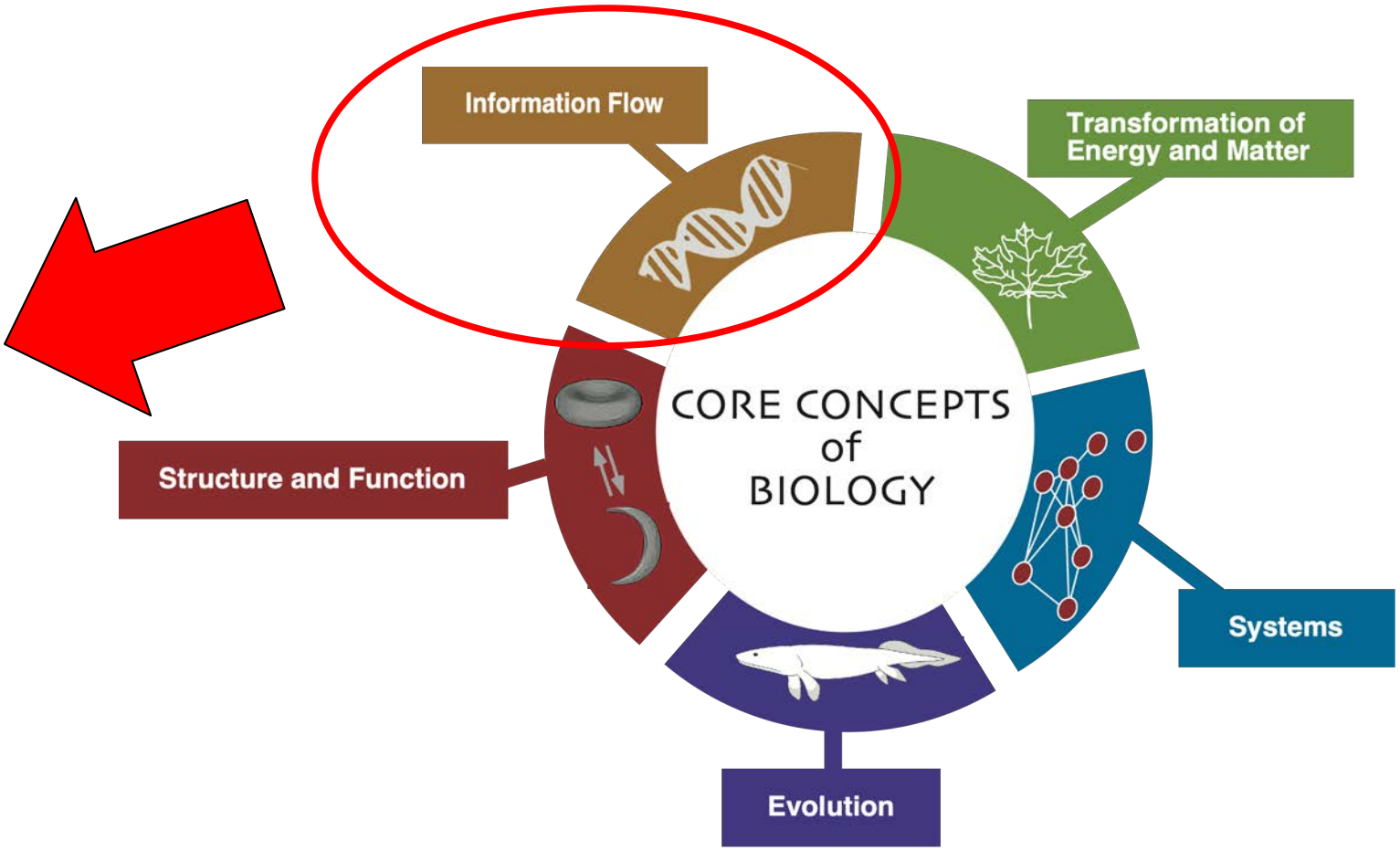
The pastor at your church is asking you about what you are studying in college. When they find our you're taking a biology class they ask you...

"Have you heard of "savior siblings? I heard that scientists and doctors can help families use IVF and genetic testing to make embryos. Then, somehow a new baby could save an older sibling with a deadly disease."

Your pastor asks, "How does that work, and what do you think about it?"

J





Information Flow

Why should you care?!?

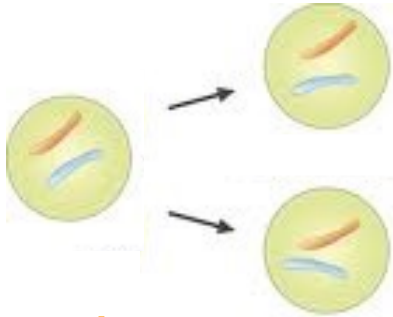
Information flow is about where you came from, how you exist as a unique person right now, and what you will leave on Earth when you are gone…

Where we've been and where we're going…

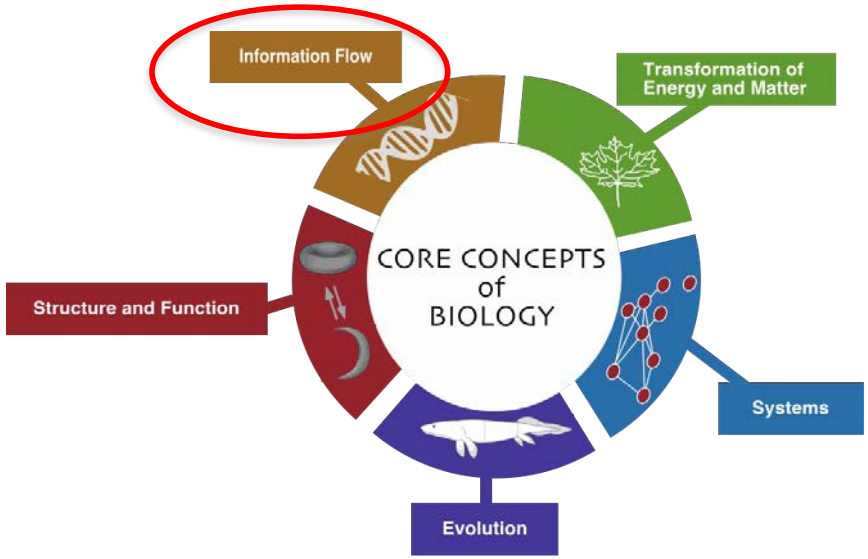
Information Flow

- Gene Expression (Central Dogma)
  - From DNA (gene) through RNA to Protein to Trait
- Cell Division (mitosis)
  - From a parent cell to two new cells

two new cells



- Inheritance (meiosis)
  - From one generation to the next generation





Information Flow

First…think on your own…



Do you think your cousin should worry?

What could you tell your cousin about the chance of passing the cancer to her biological children?

What other information would you like to know to help your cousin?

Information Flow

Clicker Question



Do you think your cousin should worry?

A. Yes, and I can explain why.
B. Yes, but I don't have an explanation.
C. No, and I can explain why.
D. No, but I don't have an explanation.
E. I don't know.

Information Flow

Now…share with your shoulder buddy or as a team of three…



Do you think your cousin should worry?

What could you tell your cousin about the chance of passing the cancer to her biological children?

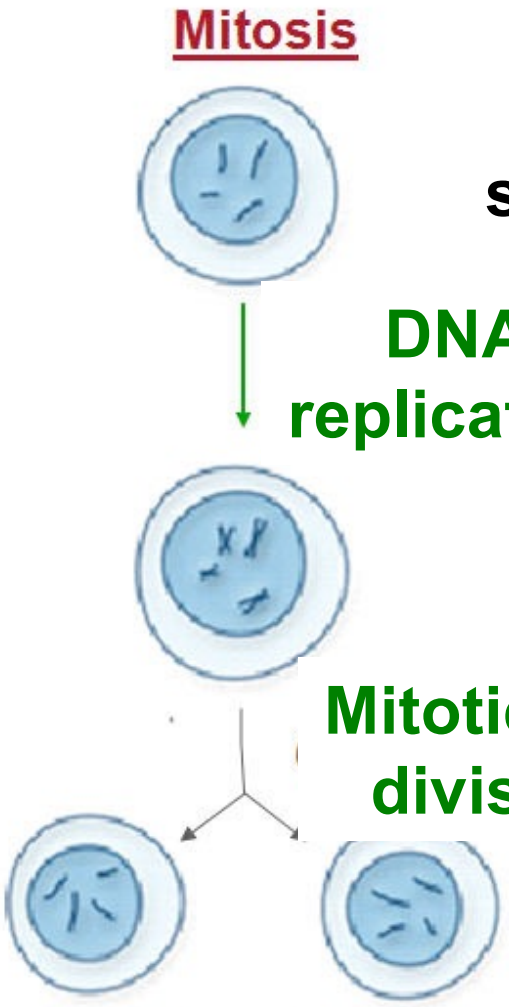
What other information would you like to know to help your cousin?

Person with the longest hair starts



Remember, cancer is unregulated cell division…

Key Idea 1: Mitosis occurs in somatic cells and produces two identical cells

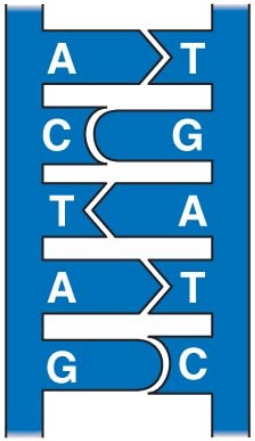


Somatic cell = all cells that are NOT in the germline making eggs/sperm

DNA Replication

DNA replication **is the process of producing two identical copies from one original DNA molecule.**

Before a cell divides (mitosis), the DNA must be copied (replicated).



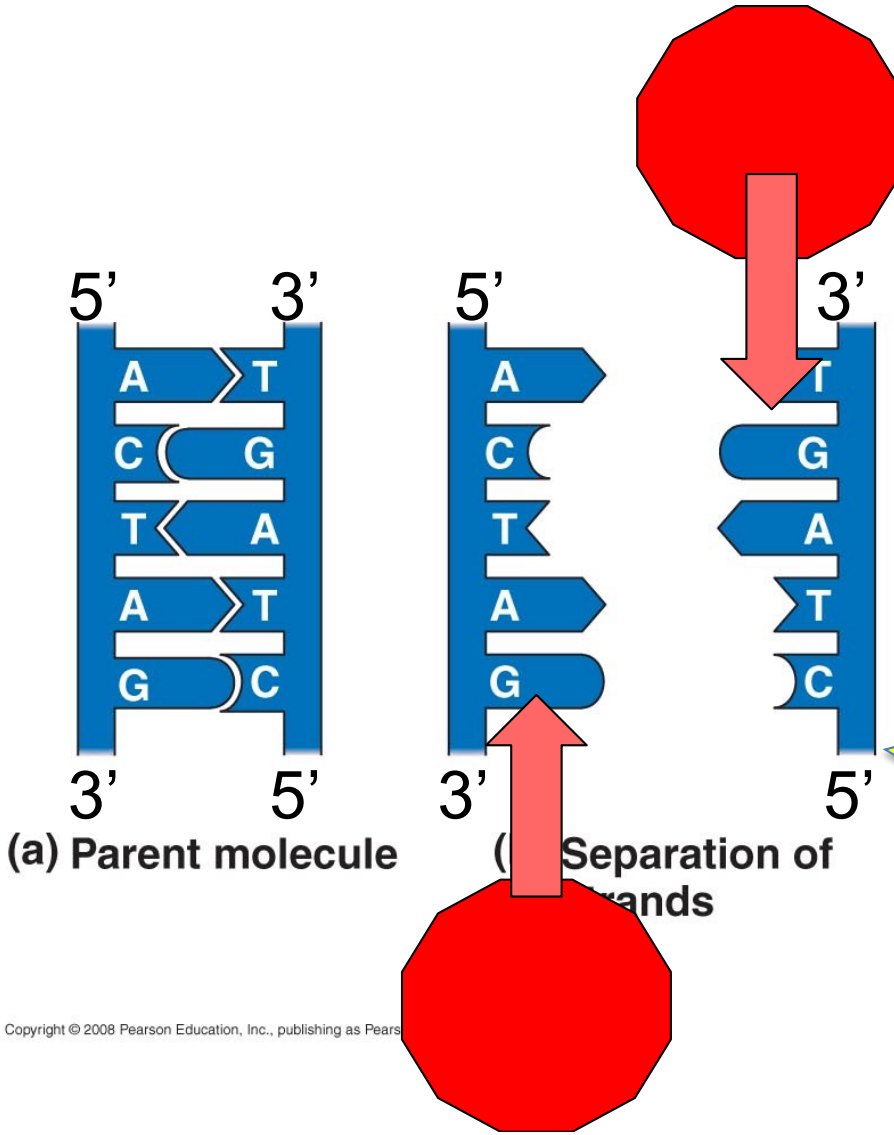
(a) Parent molecule

What is the name of the protein whose job is to make a copy of a DNA molecule?!?

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DNA Replication

DNA Polymerase
"DNA Builder"

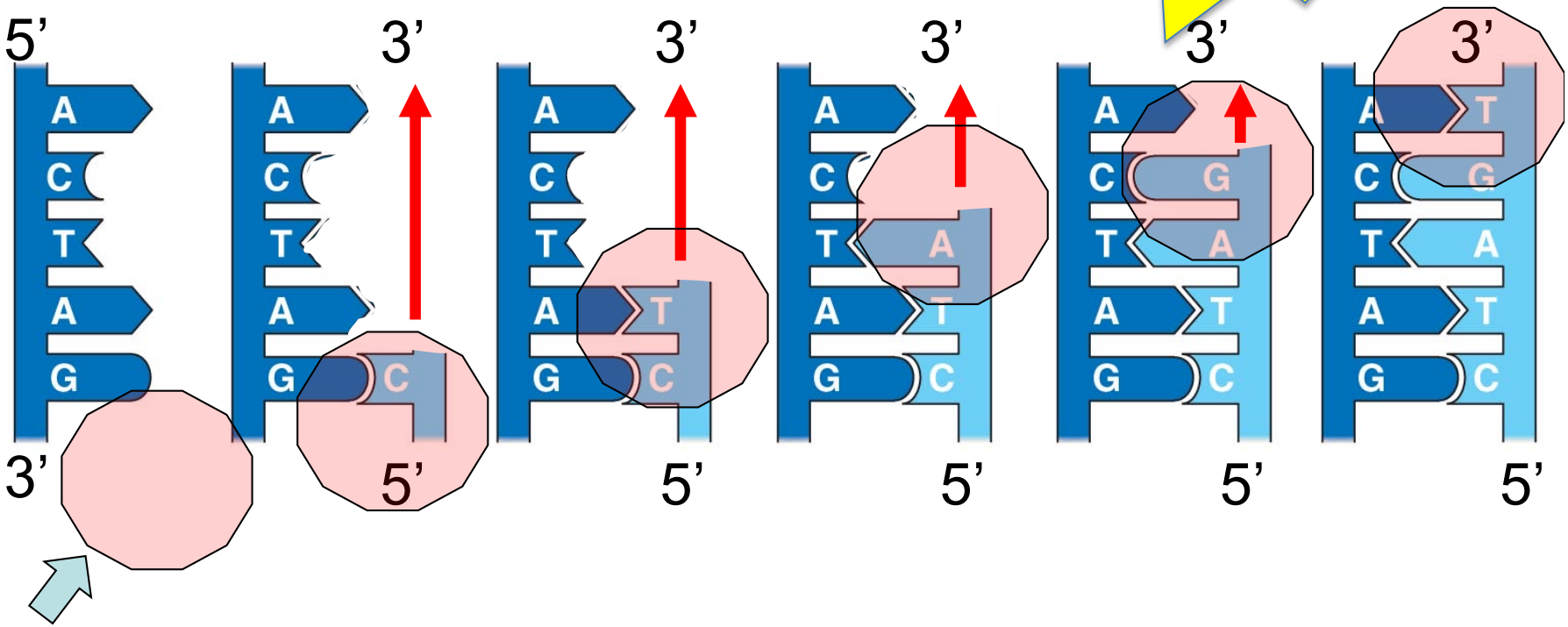


A pairs with T
G pairs with C

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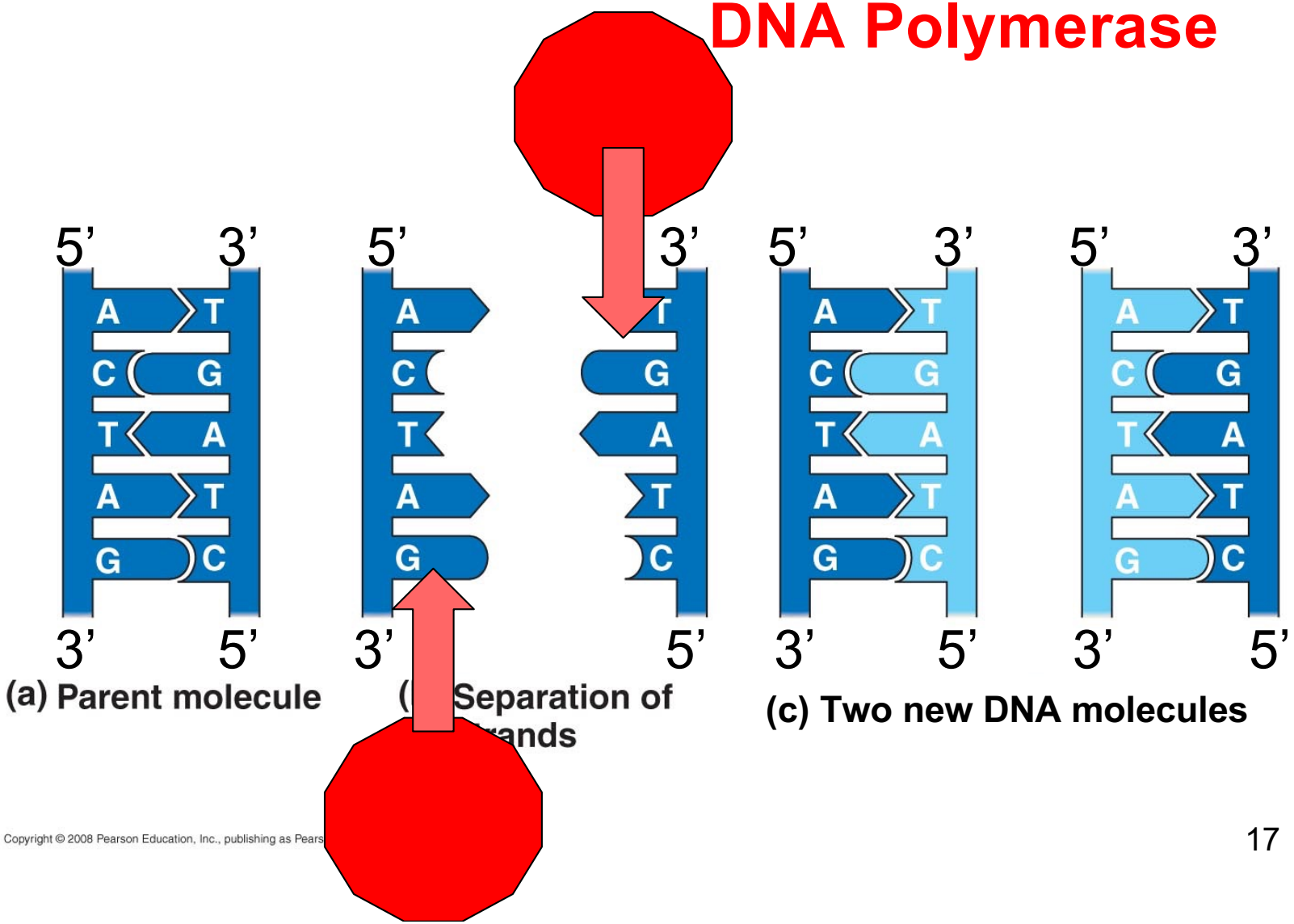
Let's watch one DNA strand be copied

A pairs with T
G pairs with C



DNA Polymerase

DNA Replication



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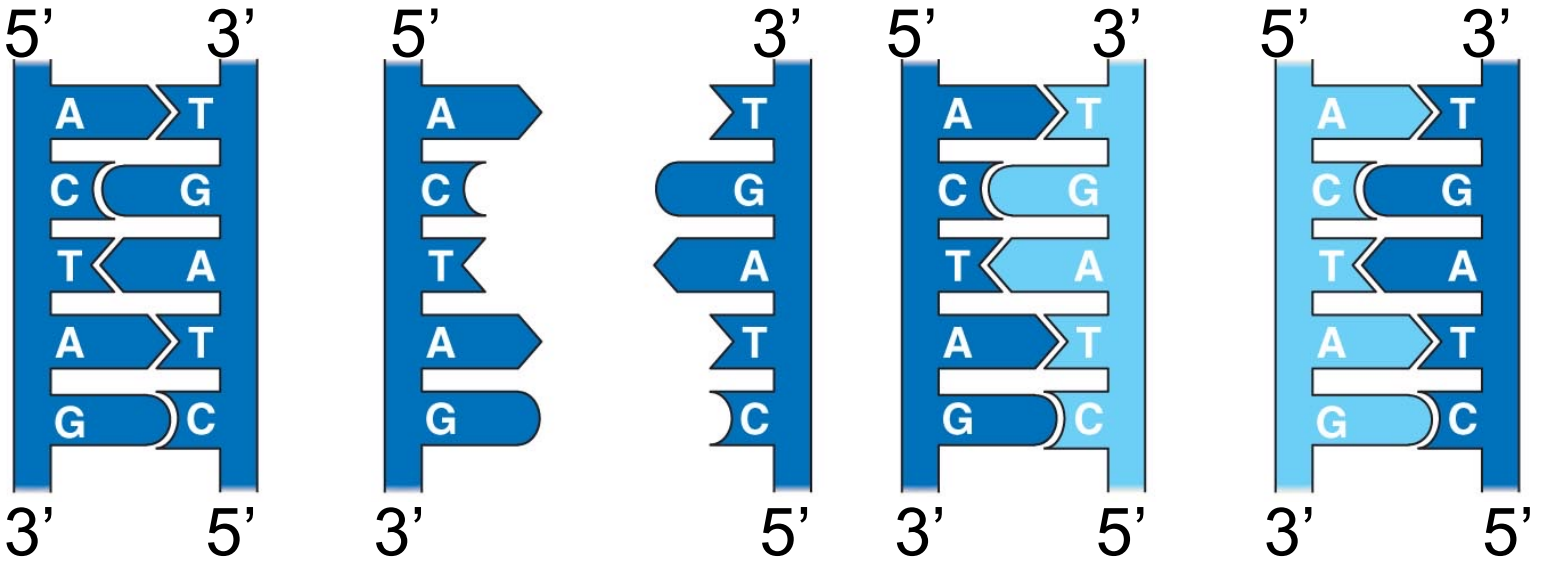
You played the role of DNA polymerase with our mystery gene puzzles…

Mystery Gene! – "Normal" Allele (form)		
Predicted complementary DNA sequence	5'	3'
TEMPLATE DNA sequence	3'TACCAAGTAGACTGAGGACTTCTT…	5'
Predicted mRNA sequence	5'	3'
Predicted PROTEIN (amino acid) sequence		

Did you copy the DNA without any mistakes?

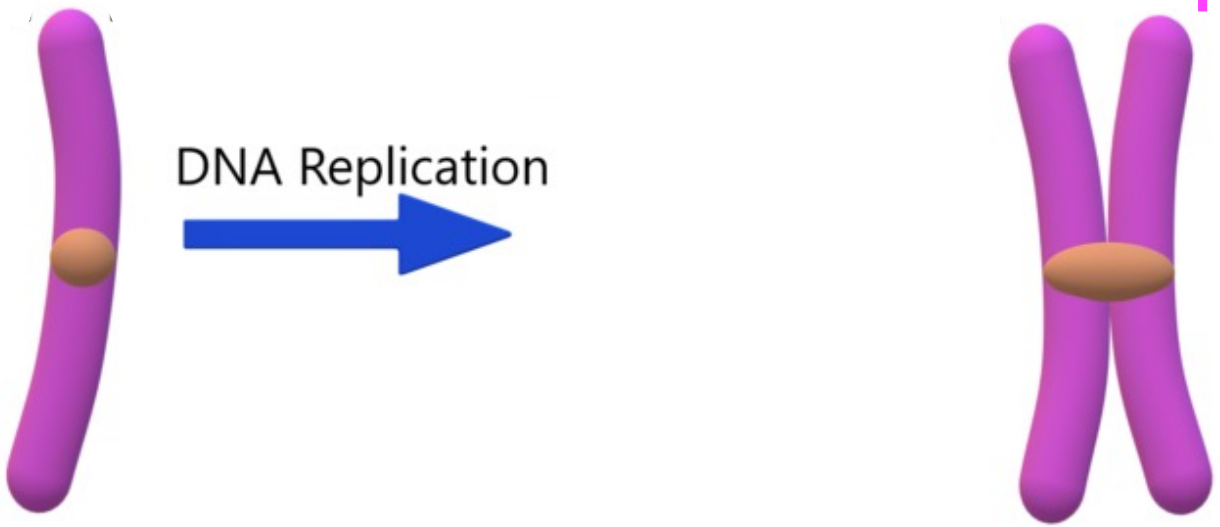
Yes, DNA polymerase can make mistakes (rare!).
One source of mutations!

DNA Replication

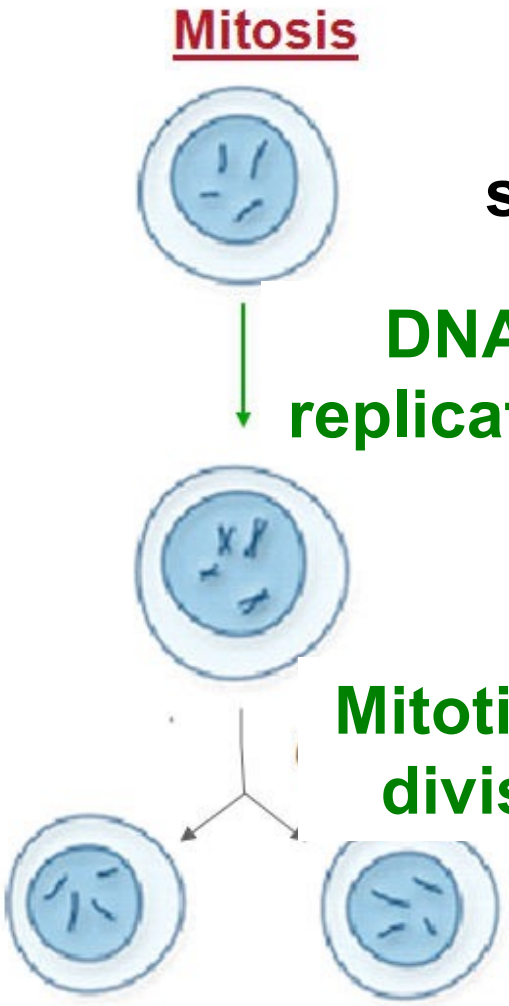


Represents 1 DNA duplex

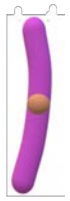
Represents 2 DNA duplexes
Identical copies

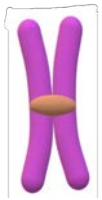


Key Idea 1: Mitosis occurs in somatic cells and produces two identical cells

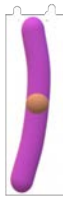


somatic cell





identical cells are produced

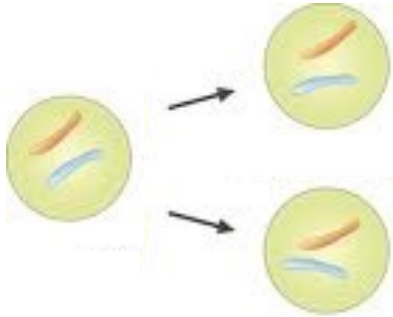


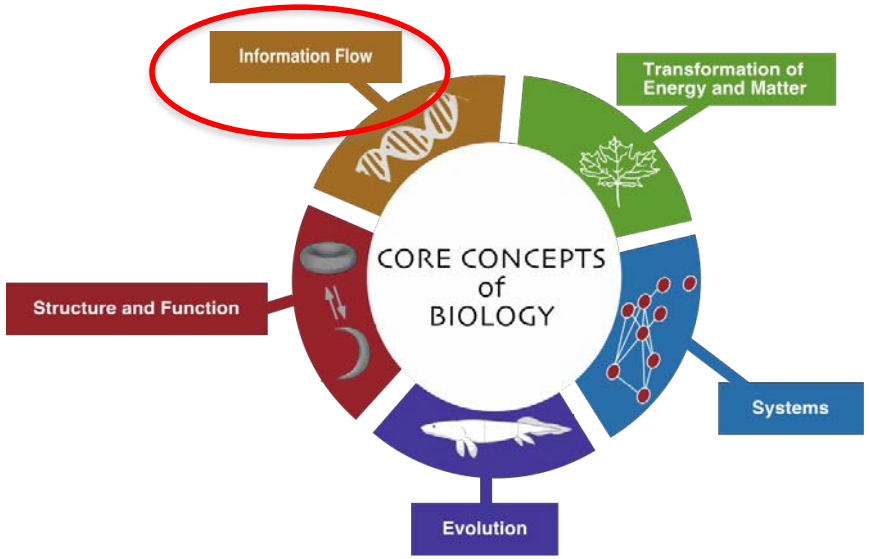
Where we've been and where we're going…

Information Flow

- Gene Expression (Central Dogma)
  - From DNA (gene) through RNA to Protein to Trait
- Cell Division (mitosis)
  - From a parent cell to two new cells

two new cells







“Peanut butter can cause cancer”

The Peanut Files Homework

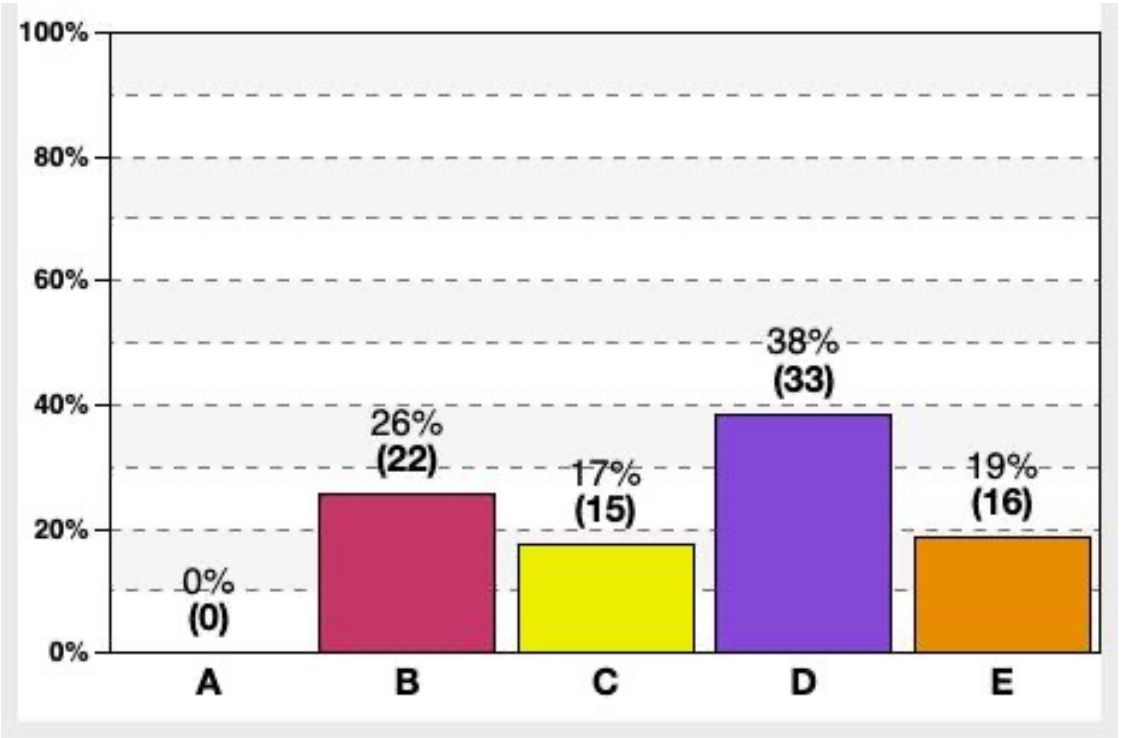
What are the potential links between peanuts and liver cancer?

What is the evidence for and against this statement?



Reflective Journal #5





Choose the answer that best describes your response to the statement:

“Peanut butter can cause cancer.”

A. Yes, peanut butter is always linked to cancer in humans.
B. Yes, peanut butter can be linked to cancer in humans.
C. No, peanut butter is never linked to cancer in humans.
D. I have no idea, and I love peanut butter.
E. I have no idea, and I hate peanut butter.

Ask questions

Be skeptical

Use evidence

"butter can ca[illegible]e cancer"

The Peanut Files H[illegible]ork

What are the potential links [illegible]een peanuts and liver cancer:

What is the evidence for and against this statement?





Revised
Clicker Question
Click in!

Choose the answer that best describes your response to the statement:

"Peanut butter can cause cancer."

A. Yes, peanut butter always causes cancer.

B. Yes, peanut butter can cause cancer under certain circumstances.

C. No, peanut butter cannot ever cause cancer.

D. I didn't do The Peanut Files homework, so I don't really know…

E. I did The Peanut Files homework, and I'm still really confused!

Pair Discussion

with your shoulder buddy or your team of three:

"Peanut butter can cause cancer"

The Peanut Files Homework

What are the potential links between peanuts and liver cancer?

What is the evidence for and against this statement?

Check your information with a neighbor!

Be prepared to share out one idea!



“Peanut butter can cause cancer”

The Peanut Files Homework

What are the potential links between peanuts and liver cancer?

What is the evidence for or against this statement?

(any specific genes mentioned?)

- …?
- …?
- …?
- …?
- …?
- …?

See Canvas for in-class responses



Case Study #1: The Altoids Scare



My best friend and I were driving to lunch when I asked her for a breathmint. She offered me some gum but said, "I don't have any breathmints after the Altoids scare."

"Give me a break, what Altoids scare?"

"I read on the internet that you can get cancer from Altoids. After all, they are from Britain, and they do have gelatin in them."

I know I've read some stuff about Mad Cow disease. I let the whole day go by without challenging my friend on her ideas. I just had so many questions that I didn't know the answers to…

May sound similar to The Peanut Files, but different… because little to no evidence for Altoids Scare…

“Peanut butter can cause cancer”

The Peanut Files Homework

So how could peanut butter possibly be connected to a mutation in a human cell?

What would have to happen between eating peanut butter and a DNA mutation?





Name Tent Reflection

Please THINK and WRITE (in a complete sentence) for today…

What would have to happen between eating peanut butter and a DNA mutation?

_____

_____

Your Preferred First Name

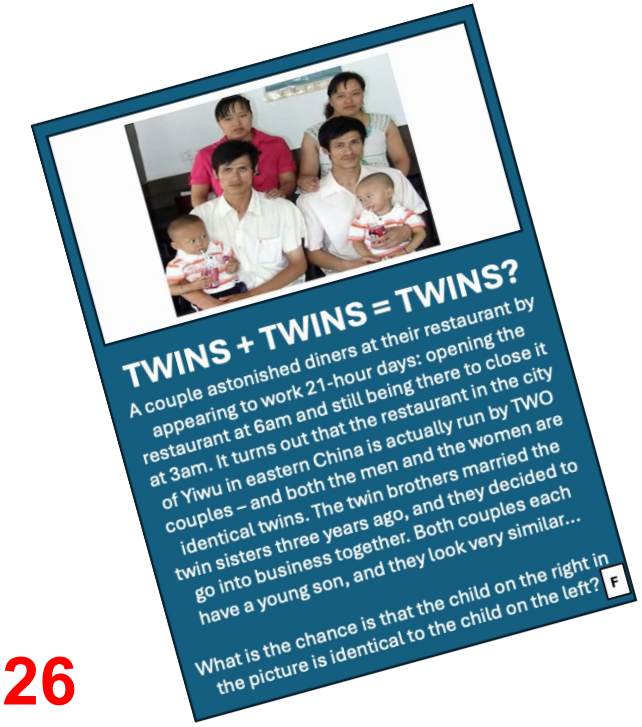
Please leave all name tents in the bin at your table after EVERY class session!

Assignments

Due Sunday, September 27, 2026 by Midnight (11:59pm)

- **Complete Reflective Journal #6 Herman and Marisa AND Twins+Twins=Twins?**

Go Team Bio 100!



Due Saturday, September 26, 2026 by Midnight (11:59pm)

- **Self-Assessment Quiz #4: Mutations and Chance of Inheriting Cancer Mutations**